JITHIN SHAJI

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Career Objective

Results-driven Robotics and Automation Engineering student with proven expertise in intelligent systems, real-time control, and autonomous navigation. Lead technical development for Team Equinox, achieving All India Rank 7th in the aBAJA competition. Eager to apply advanced technical skills to contribute to innovative and future-ready robotics and AI engineering projects.

Education

Bachelor of Technology (B.Tech) in Robotics and Automation – Saintgits College of Engineering, Kottayam
CGPA: 7.3 | Expected Graduation: 2026

Higher Secondary Education (Class XII) – Good Shepherd Public School
Percentage: 85% | Year of Completion: 2022

Higher Secondary Education (Class X) – Good Shepherd Public School
Percentage: 85% | Year of Completion: 2020

Skills

- **Languages:** ROS, Python, C++
- **Robotics Software:** Gazebo, Navigation, SLAM, PID Control, URDF/XACRO
- **Personal Attributes:** Leadership ,Teamwork ,Communication , Self-Learning Ability ,Time Management
- **Hardware & Comms:** Raspberry Pi, ESP32, Arduino, Microcontrollers, LIDAR, IMU, Ultrasonic, IR, CAN bus, UART, I2C, Blynk IoT
- **Computer Vision & AI**: OpenCV, Flask, Machine Learning Concepts, Face Recognition
- **Tools & Platforms**: Linux, Git, GitHub, SolidWorks, AutoCAD, Fusion 360,Kicad
- **Languages (Spoken/Written):** English, Malayalam

Professional & Technical Experience

Team Equinox (ABAJA SAEINDIA 2025)
System Integration Lead | Saintgits College of Engineering, Kerala
Mar 2024 – Present

- Led technical development for Team Equinox, achieving **All India Rank 7th** in the **aBAJA SAE INDIA 2025** national competition.
- Developed a Level 2 autonomous buggy, integrating sensors, path planning, and CAN-based communication, along with system-level integration and implementation of PID and Stanley control algorithms for perception, control, and navigation subsystems.
- Handled electrical system design and wiring, including power distribution, sensor interfacing, and communication line optimization for reliability and safety.

Internships

1. Industrial Robot Programming Intern , PSG – FANUC Centre for Advanced CNC and Robotics (FANUC Authorised Training Centre)
May 2024 – June 2024

- Programmed industrial FANUC robots, gaining hands-on experience in robot motion control and automation applications.
- Learned industrial robot programming concepts and operations using FANUC robots and Gained hands-on experience in robot motion control and automation applications.

2. Artificial Intelligence & Machine Learning Intern, Innovate Intern
Jan 2024 – Feb 2024

- Designed and developed a Flask-based Face Recognition Attendance System with a GUI interface during a 4-week AI/ML internship.

3. ROS 2 Training Participant , PSG – FANUC Centre for Advanced CNC and Robotics (FANUC Authorised Training Centre)
May 2025 – June 2025

- Acquired foundational skills in ROS 2 architecture, nodes, topics, and communication mechanisms.
- Developed and tested basic robot control applications using ROS 2 framework.

4. Embedded Systems and Robotics Intern, Techmaghi
June 2024 – July 2024

- Developed and coded a functional mobile robot, implementing real-time motion control and obstacle detection systems.

Selected Technical Projects

Swarm Robot Development (ROS 2) Completed
Role: ROS 2 Communication & Simulation Lead

- Developing a multi-robot system inspired by natural swarm behavior for coordinated warehouse automation tasks.
- Implementing ROS 2 framework for inter-robot communication, formation control, and obstacle
- avoidance algorithm.
- Conducting simulations and initial hardware trials to validate decentralized decision-making and scalability.

B.R.A.V.O (Hospital Assistive Robot with Smart Triage & Autonomous Navigation) Ongoing
Role: ROS 2 Integration, Simulation & Raspberry Pi Lead

- Raspberry Pi–based robot for patient vitals collection and ROS 2–driven triage.
- Autonomous navigation using LIDAR, IMU, and mecanum wheels with room-specific smart action.
- Implemented autonomous navigation using LIDAR, IMU, and mecanum wheels, integrating ROS 2 for sensor fusion and decision-making.

VANIS Robot (Hack for Nothing Hackathon) Role: IoT Systems Integrator
Role:Lead Developer – Person Detection & Raspberry Pi Integration Ongoing

- Integrated phone camera (as IP camera), ultrasonic, and IR sensors for vision, obstacle avoidance, and initiating autonomous hiding behavior.
- Implemented decision-making logic based on face detection to trigger hiding maneuvers, shadow detection.
- Developed a Raspberry Pi–based robotic system capable of detecting human faces using OpenCV and

Smart Traffic & Parking Systems
Hackathon Projects – Revathon (Chengannur Government College of Engineering Completed
Role: Lead Developer and Designer

- Designed Arduino-based smart traffic light and automated parking systems using LEDs, ultrasonic sensors, and servo motors for efficient traffic and parkingmanagement.
- Created precise 3D CAD models from 2D images using Fusion 360 for engineering design applications.

Leadership & Achievements

- **BAJA SAEINDIA National Competition (2025):** Secured All India Rank 7th as System Integration Lead for Team Equinox.
- **Venue Coordination Head** – 11th National Level Technical Project Exhibition “Srishti 2025” : Directed venue logistics and coordination for a premier national-level technical exhibition, managing scheduling and on-site operations for 50+ participating teams.
- **Sub-Head - Event “County Cricket,” Nakshatra 2025 (National Level Techno-Cultural Fest) (2025)**: Coordinated planning, logistics, and execution of a major event at a national-level fest, managing participant registration and on-ground operations.
- **NPTEL**: Solid Mechanics (Elite) | 2024
- **Google Project Management** – Google (Coursera) | 2024
- **Fundamentals on Robotics and Industrial Automation – L&T Edu Tech** (Coursera) | 2024
- Concluded the course " **Python Programming**" from Electronics and communication department.
- **Executive Committee Member** – College Magazine 2024 - Present :
  Contributed to planning, content review, and coordination for the college magazine's

Workshops

- Workshop on SELF DRIVING CAR conducted by SAEINDIA Southern Section at SAEISS, Guindy, Chennai.
- ROS NOETIC in action workshop on 8th & 11th april 2024 conducted by iedc@saintgits under the banner of ipl 3.0
- workshop on PCB DESIGN as part of THE ARC hosted by IEEE Student Branch Chapter

Membership in Professional Body

- **SAEINDIA: Member :** Member, Society of Automotive Engineers India
- **IEEE: Member :** Member of IEEE Student Branch
- **IETE: Member :** Member of Institution of Electronics and Telecommunication Engineers (IETE)